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```java
// 1.Write a program to take a list of 10 farmers, fruits they sell, their cities & the quantity of fruits they have as input. Save the data to notepad before & after sorting based on quantity in tabular manner. Also perform search operation based on farmers' name.

import java.io.*;
import java.util.*;

class Farmer {
    String name, fruit, city;
    int quantity;

    Farmer(String name, String fruit, String city, int quantity) {
        this.name = name;
        this.fruit = fruit;
        this.city = city;
        this.quantity = quantity;
    }
}

public class FarmerData {
    public static void main(String[] args) throws IOException {
        Scanner sc = new Scanner(System.in);
        ArrayList<Farmer> list = new ArrayList<>();

        // Input
        for (int i = 0; i < 10; i++) {
            System.out.println("Enter Farmer Name, Fruit, City, Quantity:");
            String name = sc.next();
            String fruit = sc.next();
            String city = sc.next();
            int quantity = sc.nextInt();

            list.add(new Farmer(name, fruit, city, quantity));
        }

        // Write BEFORE sorting
        BufferedWriter bw = new BufferedWriter(new FileWriter("data.txt"));
        bw.write("----- BEFORE SORTING -----\n");
        bw.write("Name\tFruit\tCity\tQuantity\n");

        for (Farmer f : list) {
            bw.write(f.name + "\t" + f.fruit + "\t" + f.city + "\t" + f.quantity + "\n");
        }

        // Sorting by quantity
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        list.sort((a, b) -> a.quantity - b.quantity);

        // Write AFTER sorting
        bw.write("\n----- AFTER SORTING -----\n");
        bw.write("Name\tFruit\tCity\tQuantity\n");

        for (Farmer f : list) {
            bw.write(f.name + "\t" + f.fruit + "\t" + f.city + "\t" + f.quantity + "\n");
        }

        bw.close();

        System.out.println("Data saved to file (data.txt)");

        // Search
        System.out.print("Enter farmer name to search: ");
        String searchName = sc.next();

        boolean found = false;

        for (Farmer f : list) {
            if (f.name.equalsIgnoreCase(searchName)) {
                System.out.println("Found:");
                System.out.println("Name: " + f.name);
                System.out.println("Fruit: " + f.fruit);
                System.out.println("City: " + f.city);
                System.out.println("Quantity: " + f.quantity);
                found = true;
            }
        }

        if (!found) {
            System.out.println("Farmer not found.");
        }

        sc.close();
    }
}

//output
Enter Farmer Name, Fruit, City, Quantity:
Ram Mango Delhi 500
Enter Farmer Name, Fruit, City, Quantity:
Shyam Apple Mumbai 1200
Enter Farmer Name, Fruit, City, Quantity:
Amit Banana Pune 300
Enter Farmer Name, Fruit, City, Quantity:
Ravi Orange Delhi 800
Enter Farmer Name, Fruit, City, Quantity:
```

```
Sita Mango Jaipur 1500
Enter Farmer Name, Fruit, City, Quantity:
Geeta Apple Indore 200
Enter Farmer Name, Fruit, City, Quantity:
Mohan Banana Bhopal 900
Enter Farmer Name, Fruit, City, Quantity:
Ramesh Mango Delhi 700
Enter Farmer Name, Fruit, City, Quantity:
Kiran Apple Pune 400
Enter Farmer Name, Fruit, City, Quantity:
Neha Orange Mumbai 600
Data saved to file (data.txt)
Enter farmer name to search: kiran
Found:
Name: Kiran
Fruit: Apple
City: Pune
Quantity: 400


//2.Write a program to take quantity & quality of fruits Mango, Apple & Papaya as input. Quality may be good, bad or average. If entered quantity of fruit is greater than 1000, store it as half. Perform search for a given fruit with given quality.

import java.util.*;

class Fruit {
    String name;
    int quantity;
    String quality;

    Fruit(String name, int quantity, String quality) {
        this.name = name;
        this.quality = quality;

        // If quantity > 1000, store half
        if (quantity > 1000) {
            this.quantity = quantity / 2;
        } else {
            this.quantity = quantity;
        }
    }

    void display() {
        System.out.println("Fruit: " + name +
                ", Quantity: " + quantity +
                ", Quality: " + quality);
    }
}

```

```java
public class FruitProgram {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Fruit[] fruits = new Fruit[3];

        String[] fruitNames = { "Mango", "Apple", "Papaya" };

        // Input
        for (int i = 0; i < 3; i++) {
            System.out.println("Enter quantity for " + fruitNames[i] + ": ");
            int qty = sc.nextInt();

            System.out.println("Enter quality (good/bad/average): ");
            String quality = sc.next();

            fruits[i] = new Fruit(fruitNames[i], qty, quality);
        }

        // Display all data
        System.out.println("\nStored Data:");
        for (Fruit f : fruits) {
            f.display();
        }

        // Search
        System.out.println("\nEnter fruit name to search:");
        String searchFruit = sc.next();

        System.out.println("Enter quality to search:");
        String searchQuality = sc.next();

        boolean found = false;

        for (Fruit f : fruits) {
            if (f.name.equalsIgnoreCase(searchFruit) &&
                    f.quality.equalsIgnoreCase(searchQuality)) {

                System.out.println("\nFruit Found:");
                f.display();
                found = true;
            }
        }

        if (!found) {
            System.out.println("\nNo matching fruit found.");
        }

        sc.close();
    }
```

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}

//output
Enter quantity for Mango:
1600
Enter quality (good/bad/average):
good
Enter quantity for Apple:
500
Enter quality (good/bad/average):
bad
Enter quantity for Papaya:
133
Enter quality (good/bad/average):
good

Stored Data:
Fruit: Mango, Quantity: 800, Quality: good
Fruit: Apple, Quantity: 500, Quality: bad
Fruit: Papaya, Quantity: 133, Quality: good

Enter fruit name to search:
papaya
Enter quality to search:
good

Fruit Found:
Fruit: Papaya, Quantity: 133, Quality: good

//3.There are four orchards, namely A, B, C & D with fruits in them as Apple, Litchi, Mango, Apple respectively. The qualities of the fruits in the orchards are good, good, good & bad respectively.
The available orchards in a tree-view.
Search for a given fruit in the available orchards after taking input from the user & print the tree.

import java.util.*;

class Orchard {
    String name;
    String fruit;
    String quality;

    Orchard(String name, String fruit, String quality) {
        this.name = name;
        this.fruit = fruit;
        this.quality = quality;
    }
}

```

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public class OrchardTree {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Predefined orchards
        Orchard[] orchards = {
                new Orchard("A", "Apple", "Good"),
                new Orchard("B", "Litchi", "Good"),
                new Orchard("C", "Mango", "Good"),
                new Orchard("D", "Apple", "Bad")
        };

        // Input for search
        System.out.print("Enter fruit to search: ");
        String searchFruit = sc.next();

        boolean found = false;

        System.out.println("\nTree View:\n");
        System.out.println("Orchards");

        for (Orchard o : orchards) {
            if (o.fruit.equalsIgnoreCase(searchFruit)) {
                System.out.println("├── " + o.name + " (" + o.fruit + ", " + o.quality + ")  <-- Found");
                found = true;
            } else {
                System.out.println("├── " + o.name + " (" + o.fruit + ", " + o.quality + ")");
            }
        }

        if (!found) {
            System.out.println("\nFruit not found in any orchard.");
        }

        sc.close();
    }
}

//Output
Enter fruit to search: litchi

Tree View:

Orchards
â??â??â?? A (Apple, Good)
â??â??â?? B (Litchi, Good)  <-- Found
â??â??â?? C (Mango, Good)
â??â??â?? D (Apple, Bad)
```

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//4.Take a map provided as a graph & display all the subtrees of connected points.

import java.util.*;

public class GraphSubtrees {

    static void dfs(int node, boolean[] visited, ArrayList<ArrayList<Integer>> graph, List<Integer> component) {
        visited[node] = true;
        component.add(node);

        for (int neighbor : graph.get(node)) {
            if (!visited[neighbor]) {
                dfs(neighbor, visited, graph, component);
            }
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of vertices: ");
        int n = sc.nextInt();

        System.out.print("Enter number of edges: ");
        int e = sc.nextInt();

        ArrayList<ArrayList<Integer>> graph = new ArrayList<>();

        for (int i = 0; i < n; i++)
            graph.add(new ArrayList<>());

        // Input edges
        System.out.println("Enter edges (u v):");
        for (int i = 0; i < e; i++) {
            int u = sc.nextInt();
            int v = sc.nextInt();

            graph.get(u).add(v);
            graph.get(v).add(u); // undirected graph
        }

        boolean[] visited = new boolean[n];

        System.out.println("\nConnected Subtrees:");

        for (int i = 0; i < n; i++) {
            if (!visited[i]) {
                List<Integer> component = new ArrayList<>();
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                dfs(i, visited, graph, component);

                System.out.println(component);
            }
        }

        sc.close();
    }
}

//Output

Enter number of vertices: 9
Enter number of edges: 7
Enter edges (u v):
3 4
1 3
2 4
5 6
4 5
3 6
1 4

Connected Subtrees:
[0]
[1, 3, 4, 2, 5, 6]
[7]
[8]

//5.Enter entry point & goal point & produce the matching subtree of the graph.

import java.util.*;

public class GraphPath {

    static boolean dfs(int current, int goal, boolean[] visited,
            ArrayList<ArrayList<Integer>> graph,
            List<Integer> path) {

        visited[current] = true;
        path.add(current);

        if (current == goal)
            return true;

        for (int neighbor : graph.get(current)) {
            if (!visited[neighbor]) {
                if (dfs(neighbor, goal, visited, graph, path))
                    return true;
            }
```

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        }

        // Backtrack if path not found
        path.remove(path.size() - 1);
        return false;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of vertices: ");
        int n = sc.nextInt();

        System.out.print("Enter number of edges: ");
        int e = sc.nextInt();

        ArrayList<ArrayList<Integer>> graph = new ArrayList<>();

        for (int i = 0; i < n; i++)
            graph.add(new ArrayList<>());

        System.out.println("Enter edges (u v):");
        for (int i = 0; i < e; i++) {
            int u = sc.nextInt();
            int v = sc.nextInt();

            graph.get(u).add(v);
            graph.get(v).add(u);
        }

        System.out.print("Enter entry point: ");
        int start = sc.nextInt();

        System.out.print("Enter goal point: ");
        int goal = sc.nextInt();

        boolean[] visited = new boolean[n];
        List<Integer> path = new ArrayList<>();

        if (dfs(start, goal, visited, graph, path)) {
            System.out.println("Matching Subtree (Path): " + path);
        } else {
            System.out.println("No path exists between given points.");
        }

        sc.close();
    }
}

//output
```

```

Enter number of vertices: 9
Enter number of edges: 7
Enter edges (u v):
1 2
1 3
2 4
4 5
5 3
1 6
4 3
Enter entry point: 1
Enter goal point: 4
Matching Subtree (Path): [1, 2, 4]

//6.Give the weightage for each edge of nodes, and produce the shortest path from start to end node.

import java.util.*;

public class ShortestPath {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, e;
        System.out.print("Enter number of vertices: ");
        n = sc.nextInt();

        System.out.print("Enter number of edges: ");
        e = sc.nextInt();

        int[][] graph = new int[n][n];

        // Initialize with large value
        for (int i = 0; i < n; i++)
            Arrays.fill(graph[i], 9999);

        System.out.println("Enter edges (u v weight):");
        for (int i = 0; i < e; i++) {
            int u = sc.nextInt();
            int v = sc.nextInt();
            int w = sc.nextInt();

            graph[u][v] = w;
            graph[v][u] = w;
        }

        System.out.print("Enter start node: ");
        int start = sc.nextInt();
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        System.out.print("Enter end node: ");
        int end = sc.nextInt();

        int[] dist = new int[n];
        boolean[] visited = new boolean[n];
        int[] parent = new int[n];

        Arrays.fill(dist, 9999);
        Arrays.fill(parent, -1);

        dist[start] = 0;

        // Simple Dijkstra (without priority queue)
        for (int i = 0; i < n - 1; i++) {

            int min = 9999, u = -1;

            for (int j = 0; j < n; j++) {
                if (!visited[j] && dist[j] < min) {
                    min = dist[j];
                    u = j;
                }
            }

            visited[u] = true;

            for (int v = 0; v < n; v++) {
                if (graph[u][v] != 9999 && !visited[v]) {
                    if (dist[u] + graph[u][v] < dist[v]) {
                        dist[v] = dist[u] + graph[u][v];
                        parent[v] = u;
                    }
                }
            }
        }

        // Print path
        List<Integer> path = new ArrayList<>();
        for (int v = end; v != -1; v = parent[v])
            path.add(v);

        Collections.reverse(path);

        System.out.println("Shortest Distance: " + dist[end]);
        System.out.println("Path: " + path);

        sc.close();
    }
}
```

```

//output

Enter number of vertices: 5
Enter number of edges: 4
Enter edges (u v weight):
1 3 2
1 4 2
1 2 3
1 3 4
Enter start node: 1
Enter end node: 3
Shortest Distance: 4
Path: [1, 3]

//7.Write a program to demonstrate the use of all keywords made for exception handling.

import java.util.*;

public class ExceptionDemo {

    // using 'throws'
    static void checkNumber(int num) throws Exception {
        if (num < 0) {
            // using 'throw'
            throw new Exception("Number cannot be negative");
        } else {
            System.out.println("Valid number: " + num);
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try { // using 'try'

            System.out.print("Enter a number: ");
            int num = sc.nextInt();

            int result = 10 / num; // may cause ArithmeticException
            System.out.println("Result: " + result);

            checkNumber(num); // method that uses throws

        } catch (ArithmeticException e) { // using 'catch'
            System.out.println("Error: Cannot divide by zero");

        } catch (Exception e) { // general exception
            System.out.println("Exception: " + e.getMessage());
```

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        } finally { // using 'finally'
            System.out.println("Program finished (finally block executed)");
        }

        sc.close();
    }
}

//Output

Enter a number: -3
Result: -3
Exception: Number cannot be negative
Program finished (finally block executed)

//8.Write a program to withraw money from a bank account, if the withdrawn money is greater than the balance, then it throws NoMoneyException.

import java.util.*;

// Custom Exception
class NoMoneyException extends Exception {
    NoMoneyException(String message) {
        super(message);
    }
}

public class BankAccount {

    static void withdraw(int balance, int amount) throws NoMoneyException {
        if (amount > balance) {
            throw new NoMoneyException("Insufficient Balance!");
        } else {
            balance -= amount;
            System.out.println("Withdrawal successful");
            System.out.println("Remaining Balance: " + balance);
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter account balance: ");
        int balance = sc.nextInt();

        System.out.print("Enter amount to withdraw: ");
        int amount = sc.nextInt();

        try {
```

```
            withdraw(balance, amount);
        } catch (NoMoneyException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        sc.close();
    }
}

//Output

Enter account balance: 5000
Enter amount to withdraw: 600
Withdrawal successful
Remaining Balance: 4400

Enter account balance: 500
Enter amount to withdraw: 1000
Exception: Insufficient Balance!

//9.Write a program which enables to use a platform by a train one at a time.

class Platform {

    synchronized void usePlatform(String trainName) {
        System.out.println(trainName + " is entering the platform");

        try {
            Thread.sleep(2000); // simulate time on platform
        } catch (InterruptedException e) {
            System.out.println(e);
        }

        System.out.println(trainName + " is leaving the platform\n");
    }
}

class Train extends Thread {
    Platform p;
    String name;

    Train(Platform p, String name) {
        this.p = p;
        this.name = name;
    }

    public void run() {
        p.usePlatform(name);
    }
}
```

```java

public class RailwayStation {
    public static void main(String[] args) {

        Platform platform = new Platform();

        Train t1 = new Train(platform, "Train A");
        Train t2 = new Train(platform, "Train B");
        Train t3 = new Train(platform, "Train C");

        t1.start();
        t2.start();
        t3.start();
    }
}

//Output

Train A is entering the platform
Train A is leaving the platform

Train C is entering the platform
Train C is leaving the platform

Train B is entering the platform
Train B is leaving the platform

//10.Write a program of Restaurant and Waiter, which accept order if kitchen if free and chef deliver orders if he is free.

class Restaurant {

    boolean orderReady = false;


    synchronized void placeOrder(String order) {
        try {
            while (orderReady) {
                wait(); // wait if kitchen is busy
            }

            System.out.println("Waiter placed order: " + order);
            orderReady = true;

            notify(); // notify chef

        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
```

```java

    // Chef preparing order
    synchronized void prepareOrder() {
        try {
            while (!orderReady) {
                wait(); // wait if no order
            }

            System.out.println("Chef is preparing order...");
            Thread.sleep(2000); // simulate cooking

            System.out.println("Chef delivered the order\n");
            orderReady = false;

            notify(); // notify waiter

        } catch (InterruptedException e) {
            System.out.println(e);
        }
    }
}

class Waiter extends Thread {
    Restaurant r;

    Waiter(Restaurant r) {
        this.r = r;
    }

    public void run() {
        String[] orders = { "Pie", "Rice", "Pasta" };

        for (String order : orders) {
            r.placeOrder(order);
        }
    }
}

class Chef extends Thread {
    Restaurant r;

    Chef(Restaurant r) {
        this.r = r;
    }

    public void run() {
        for (int i = 0; i < 3; i++) {
            r.prepareOrder();
        }
    }
```

```
}

public class RestaurantDemo {
    public static void main(String[] args) {

        Restaurant r = new Restaurant();

        Waiter w = new Waiter(r);
        Chef c = new Chef(r);

        w.start();
        c.start();
    }
}

//Output

Waiter placed order: Pie
Chef is preparing order...
Chef delivered the order

Waiter placed order: Rice
Chef is preparing order...
Chef delivered the order

Waiter placed order: Pasta
Chef is preparing order...
Chef delivered the order
```